

Sandeep Kumar Sahu

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Education

Veer Surendra Sai University of Technology (VSSUT), Burla <i>B.Tech in Computer Science & Engineering</i>	2024 – 2028 CGPA: 8.63/10
SSVM NK Nagar, Berhampur – Class XII (CHSE): 90% Class X (BSE): 91.16%	

Experience

AI & Machine Learning Intern <i>IBM SkillsBuild – Edunet Foundation – AICTE</i>	Dec 2025 – Jan 2026
- Implemented machine learning classification models using Python and Scikit-learn on structured datasets. - Evaluated model performance using accuracy, F1-score and validation metrics.	
Cybersecurity with Generative AI Intern <i>VOIS – Edunet Foundation – AICTE</i>	Dec 2025 – Jan 2026
- Studied cybersecurity threat detection, vulnerability analysis and security monitoring techniques. - Explored Generative AI applications for cybersecurity analysis and threat awareness.	

Projects

AI-Powered Digital Safety Platform <i>AWS Student Community Day 2026 – 2nd Prize</i>	<i>AWS Lambda, API Gateway, ML</i>
- Developed a cybersecurity platform for phishing detection, spam/message detection, password-breach checking and AI-assisted security guidance. - Built serverless components using AWS Lambda and API Gateway with integrated machine learning models and security services.	
ElderCare+ <i>Hack Odisha 2025 – 7th Place</i>	<i>React, TypeScript, TanStack Query, Leaflet.js</i>
- Developed a healthcare platform for elderly users with medicine reminders, appointment scheduling and essential support services. - Integrated map-based functionality to locate nearby pharmacies, hospitals and healthcare resources.	
Network Intrusion Detection System	<i>Python, Pandas, NumPy, Scikit-learn</i>
- Developed an ML-based system to classify malicious network traffic using NSL-KDD and CICIDS2017 datasets with preprocessing and feature engineering. - Implemented Random Forest and SVM classifiers and evaluated performance using accuracy, precision, recall, F1-score and confusion matrix.	
Fake News Detection using Machine Learning	<i>Python, NLP, Scikit-learn, TF-IDF</i>
- Built an NLP classifier using the WELFake dataset with text preprocessing and TF-IDF vectorization. - Compared multiple classification algorithms using accuracy, precision, recall and cross-validation.	
Airy Forecast – Real-Time Air Quality Web App	<i>React, TypeScript, Tailwind CSS, OpenWeatherMap API</i>
Built a responsive real-time air-quality monitoring application covering 50+ cities and deployed it on Vercel.	

Technical Skills

Languages: Python, JavaScript, TypeScript
Machine Learning: Data Preprocessing, Feature Engineering, Classification, Model Evaluation, NLP
Libraries & Web: NumPy, Pandas, Scikit-learn, Matplotlib, React.js, TanStack Query, Tailwind CSS, Leaflet.js
Cloud & Tools: AWS Lambda, API Gateway, Git, GitHub, Vercel, VS Code, Jupyter, Google Colab, LaTeX
Systems: Linux, Kali Linux, SSH, Command Line
Spoken Languages: English, Hindi, Odia

Leadership Experience

Social Media Team – Enigma, The Web & Coding Club of VSSUT Burla	Present
Manage social-media content and digital promotion for workshops, coding events and technical competitions.	
Social Media Team – CSE Society, VSSUT Burla	Present

- Create digital content and support online communication for technical events and student programs.

Achievements

- **2nd Prize** – Amazon Code Conquest, AWS Student Community Day 2026.
- **7th Place** – Hack Odisha 2025 Hackathon.
- Reached **Round 5** – Hackiiik Cybersecurity Hackathon.
- Participant – Smart India Hackathon 2025 and ISRO National Hackathon 2025.
- Completed OverTheWire Bandit cybersecurity challenges (Levels 0–24).

Certifications

AI & ML Fundamentals – IBM: Machine learning fundamentals, data preprocessing, supervised learning, model training and evaluation.

Cybersecurity with Generative AI – VOIS: Cybersecurity fundamentals, threat detection, vulnerability analysis and applications of Generative AI in security.

Frontend Web Development – IBM/AICTE: HTML, CSS, JavaScript, responsive web design and fundamentals of modern frontend development.

Linux Command Line & Cybersecurity – OverTheWire: Linux command-line operations, file permissions, shell utilities and practical cybersecurity challenges through Bandit Levels 0–24.